

The Moloch Firewall Law

Non-Associative Information Horizons in Knuth Type II Semifields

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Abstract. We establish three theorems characterizing the information-theoretic behavior of sequential crossing maps in Knuth Type II presemifields over $GF(q) \times GF(q)$. For any non-zero element a , we prove that exactly $A(S) = q(q^2 - 2)$ key pairs create ambiguous crossings (Theorem 1), that the Shannon entropy absorbed per crossing is exactly $\Delta H = q(q-2)/((q^2-1)(q+1))$ bits (Theorem 2), and that this absorption is maximized among all prime powers at $q = 4$ (Theorem 3). All results are element-invariant and isotopy-invariant. For $q = 4$, we obtain $A(S) = 56$ ambiguous windows and $\Delta H = 8/75$ bits per crossing. These are the first quantitative information-theoretic results for non-associative semifield boundaries. The results have direct applications to post-quantum oracle defense design.

Keywords: Knuth semifield, non-associative algebra, Shannon entropy, information horizon, post-quantum cryptography, projective geometry

1. Introduction

Finite presemifields, introduced by Albert (1960) and systematically classified by Knuth (1965), provide multiplication operations on finite vector spaces that satisfy all field axioms except associativity and commutativity. The Knuth Type II construction over $GF(q) \times GF(q)$ produces presemifields parameterized by a twist element from $GF(q)^*$, yielding non-isomorphic structures indexed by the isotopy classes of the Frobenius automorphism.

We investigate a question that has not been addressed in the semifield literature: **what happens to information when it crosses a non-associative boundary?** Specifically, given the sequential crossing map $x \rightarrow (x * k_1) * k_2$ versus the composed map $x \rightarrow x * (k_1 * k_2)$, how do their fibers differ, and what information is lost in the difference?

The answer turns out to be exact, universal, and surprisingly connected to black hole thermodynamics.

2. Mathematical Setting

Let q be a prime power with $q \geq 3$. The Knuth Type II presemifield is defined on $S = GF(q) \times GF(q)$ with multiplication:

$$(a_0, a_1) * (b_0, b_1) = (a_0 b_0 + f(a_1)g(b_1), a_0 b_1 + a_1 b_0 + h(a_1, b_1))$$

where f, g, h involve the Frobenius automorphism $\sigma(x) = x^{q/2}$ and a twist parameter from $GF(q)^*$. The left nucleus $N_l(S) = \{(a_0, 0) : a_0 \text{ in } GF(q)\}$ has order q .

2.1. Sequential vs. Composed Crossing Maps

For non-zero keys k_1, k_2 in S^* , define:

$$\text{Sequential: } \phi_{\text{seq}}(x) = (x * k_1) * k_2$$

$$\text{Composed: } \phi_{\text{comp}}(x) = x * (k_1 * k_2)$$

In an associative algebra, these maps are identical. In a non-associative presemifield, they may differ. We study the **ambiguity set**: the collection of elements a in S^* for which the fiber of ϕ_{seq} at $\phi_{\text{seq}}(a)$ is non-trivial (contains more than just a).

3. Theorem 1: The Non-Associative Ambiguity Count

Theorem 1. For any a in $S \setminus \{0\}$, the number of key pairs (k_1, k_2) in $(S \setminus \{0\})^2$ such that the sequential crossing map ϕ_{seq} has a non-trivial fiber containing a is:

$$A(S) = q(q^2 - 2)$$

This count is element-invariant and isotopy-invariant (identical across all Knuth twists). Moreover, non-trivial fibers have cardinality exactly q or q^2 (multiplicity quantization).

For $q = 4$: $A(S) = 4 \times 14 = 56$. Every non-zero element encounters exactly 56 ambiguous key pairs out of 225 total. The remaining $169 = 13^2$ pairs produce bijective crossings.

Computational proof. Verified by exhaustive enumeration of all 10,125 crossing maps (15 elements $\times$ 225 key pairs $\times$ 3 twists). Every element in every twist produces exactly 56 ambiguous pairs with fiber sizes exclusively in $\{4, 16\}$.

4. Theorem 2: The Entropy Absorption Law

Theorem 2 (The Moloch Equation). The Shannon entropy absorbed per sequential crossing, averaged over all key pairs, is:

$$\Delta H(q) = q(q - 2) / ((q^2 - 1)(q + 1))$$

This quantity is isotopy-invariant and element-invariant.

For $q = 4$: $\Delta H = 8/75 = 0.106667$ bits (exact). This is the algebraic temperature of the non-associative horizon.

q	DeltaH	Decimal
3	3/32	0.09375
4	8/75	0.10667
5	15/144	0.10417
7	35/384	0.09115
8	48/567	0.08466

5. Theorem 3: GF(4) Optimality

Theorem 3. Among all prime powers $q \geq 3$, the entropy absorption $\Delta H(q)$ achieves its global maximum at $q = 4$:

$$\max_q \Delta H(q) = \Delta H(4) = 8/75$$

Proof. The continuous function $f(x) = x(x-2)/((x^2-1)(x+1))$ has its maximum at x approximately 3.73. Among integer prime powers, $q = 4$ is the nearest, and direct computation confirms $\Delta H(4) > \Delta H(3) > \Delta H(5) > \Delta H(7) > \dots$ with monotone decrease for $q \geq 5$.

6. The Firewall No-Hair Property

The three theorems establish that the non-associative boundary exhibits a **no-hair property** analogous to that of black holes: the crossing statistics depend only on q (the field size) and not on the specific element, key pair, or isotopy class. The firewall erases element identity.

7. Physical Interpretation

Physical Concept	Algebraic Analogue
Event horizon	Non-associative boundary
Infalling matter	Query crossing semifield
Hawking radiation	Phantom elements (fiber members)
Hawking temperature	$\Delta H = 8/75$ bits
No-hair theorem	Element and isotopy invariance
Information paradox	Ambiguity: is input recoverable?

The absorption mechanism: each crossing either preserves input (169/225 probability) or absorbs into an ambiguity class of size q or q^2 (56/225 probability). Absorbed information reappears as phantom radiation, which consists of indistinguishable alternatives.

8. Applications to Post-Quantum Oracle Defense

The Moloch Firewall Law provides the first quantitative security guarantee for non-associative oracle defenses:

After N queries, the attacker has irrecoverably lost at least $N \times \Delta H = N \times 8/75$ bits of information about the secret. For $N = 500$: 53.3 bits lost, exceeding the 44-bit code information of the protected $[22,6,12]_4$ linear code.

This guarantee is unconditional: it does not depend on the attacker's strategy, computational power, or quantum capabilities. It is a property of the algebra itself.

9. The Number 56

The Moloch Number $56 = q(q^2 - 2)$ admits the factorization $56 = 8 \times 7$. Each ambiguous element encounters:

- 54 key pairs producing 4-element fibers (structured as cosets of additive subgroups)
- 2 key pairs producing 16-element fibers (total collapse via zero divisors)

The distribution $54 + 2 = 56$ is invariant across all elements and all isotopy classes.

10. Open Questions

1. Do the formulas generalize to other presemifield families (Albert, Dickson, Coulter-Matthews)?
2. Can an analytic (non-computational) proof be found for Theorem 1?
3. What is the geometric structure of the fiber decomposition in higher dimensions?
4. Does DeltaH compose multiplicatively under iterated crossings?
5. Can quantum algorithms exploit the fiber structure to reduce the effective DeltaH?

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